

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Touch Display
Manufacturer: CTOUCH
Firmware Version: N/A
Model(s): Laser air plus

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_0_3_0	10/30/2018	Added command BLUPower. Updated Input USB status. Updated Power command syntax.
1_0_1_0	9/19/2018	Changed Power command to BLU Power command syntax, to stop the serial port on the device from closing upon powering off.
1_0_0_0	12/7/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '255' or 'Broadcast'. However, DeviceID 138 and 168 are not supported on the device. If DeviceID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Laser air plus')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Laser air plus')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value None		
# AspectRatio example InterfaceName.Set('AspectRatio', None)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command BLUPower	Value 'On'	Value 'Off'	
# BLUPower example InterfaceName.Set('BLUPower', 'On')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Freeze	Value None		
# Freeze example InterfaceName.Set('Freeze', None)			
Command Input	Value 'AV' 'VGA 3' 'DisplayPort'	Value 'VGA 1' 'HDMI 1' 'OPS'	Value 'VGA 2' 'HDMI 2' 'USB'
# Input example InterfaceName.Set('Input', 'AV')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Menu' 'Down' 'Return'	Value 'Enter' 'Left' 'Exit'	Value 'Up' 'Right'

# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')		
Command Power	Value 'On'	Value 'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command Volume	Value 0 to 100 in steps of 1	
# Volume example InterfaceName.Set('Volume', 100)		

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	
BLUPower	'On'	'Off'	
# BLUPower examples InterfaceName.Update('BLUPower') Value = InterfaceName.ReadStatus('BLUPower') InterfaceName.SubscribeStatus('BLUPower', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	
ExecutiveMode	'On'	'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'AV'	'VGA 1'	'VGA 2'
	'VGA 3'	'HDMI 1'	'HDMI 2'
	'DisplayPort'	'OPS'	'USB'
# Input examples InterfaceName.Update('Input')			

<pre>Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>		
Command Power	Value 'On'	Value 'Off'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command Volume	Value 0 to 100 in steps of 1	
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>		

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

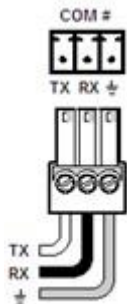
Data Bits: 8

Parity: None

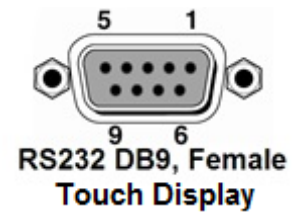
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Appendix A. Set Commands

Commands below only work when DeviceID is set to '1'.

Aspect Ratio None	\xA9\x16\x01\x81\x8A
Audio Mute Off	\xA9\x14\x01\x00\x8A
Audio Mute On	\xA9\x14\x01\x01\x8A
BLU Power On	\xA9\x12\x01\x01\x8A
BLU Power Off	\xA9\x12\x01\x00\x8A
Executive Mode Off	\xA9\x17\x01\x00\x8A
Executive Mode On	\xA9\x17\x01\x01\x8A
Freeze None	\xA9\x16\x01\xB8\x8A
Input AV	\xA9\x15\x01\x01\x8A
Input DisplayPort	\xA9\x15\x01\x07\x8A
Input HDMI 1	\xA9\x15\x01\x05\x8A
Input HDMI 2	\xA9\x15\x01\x06\x8A
Input OPS	\xA9\x15\x01\x09\x8A
Input USB	\xA9\x15\x01\x0B\x8A
Input VGA 1	\xA9\x15\x01\x14\x8A
Input VGA 2	\xA9\x15\x01\$\x8A
Input VGA 3	\xA9\x15\x014\x8A
Keypad 0	\xA9\x16\x01\xD9\x8A
Keypad 9	\xA9\x16\x01\x9E\x8A
Menu Navigation Down	\xA9\x16\x01\xD8\x8A
Menu Navigation Enter	\xA9\x16\x01\x9B\x8A
Menu Navigation Exit	\xA9\x16\x01\xD4\x8A
Menu Navigation Left	\xA9\x16\x01\x97\x8A
Menu Navigation Menu	\xA9\x16\x01\x84\x8A
Menu Navigation Return	\xA9\x16\x01\xDD\x8A
Menu Navigation Right	\xA9\x16\x01\x9F\x8A
Menu Navigation Up	\xA9\x16\x01\x92\x8A
Power Off	\xA9\x12\x01\x00\x8A
Power On	\xA9\x12\x01\x01\x8A
Volume 0	\xA9\x13\x01\x00\x8A
Volume 100	\xA9\x13\x01d\x8A

Appendix B. Update Commands

Commands below only work when DeviceID is set to '1'.

Audio Mute	\xA9\x14\x01\xAA\x8A
BLU Power	\xA9\x12\x01\xAA\x8A
Executive Mode	\xA9\x17\x01\xAA\x8A
Input	\xA9\x15\x01\xAA\x8A
Power	\xA9\x12\x01\xAA\x8A
Volume	\xA9\x13\x01\xAA\x8A